

- 1 ${}^3_1\text{H}$ contains 1 proton, 1 electron and 2 (3 – 1) neutrons. D
 ${}^3_2\text{He}$ contains 2 protons, 2 electrons and 1 (3 – 2) neutron.
- 1 ${}^3\text{He}$ has less neutrons than electrons.
2 ${}^3\text{H}$ has one proton while ${}^3\text{He}$ has two protons.
3 ${}^3\text{H}$ has two charged sub-atomic particles while ${}^3\text{He}$ has four charged sub-atomic particles.
- 2 angle of deflection $\propto \frac{\text{charge size}}{\text{mass}}$ D
- for ${}^1\text{H}^+$, $\frac{z}{m} = +\frac{1}{1}$
for ${}^2\text{X}^{2-}$, $\frac{z}{m} = -\frac{2}{2}$
Hence, angle of deflection for ${}^2\text{X}^{2-} = \underline{-4^\circ}$
- 3 A
- | | | <u>shape / angle</u> |
|---|--------------------|---------------------------------------|
| A | SF_2 | bent / $\sim 104.5^\circ$ |
| B | XeF_2 | linear / 180° |
| C | SO_3^{2-} | trigonal pyramidal / $\sim 107^\circ$ |
| D | AlH_4^- | tetrahedral / 109.5° |
- 4 The hybridisation state of N atom can be determined using similar approach as that for C atoms, except the *lone pair of electrons on N* can sometimes be occupying a sp^2 (norm) or sp^3 hybrid orbital. The question will likely provide information on the shape of the molecule with respect to N atom to link it to the hybridisation state – planar (sp^2) and tetrahedral (sp^3). D
- A is correct. All C atoms in **F** are sp^3 hybridised. Hence, each C–C bond is a σ bond formed by a sp^3 – sp^3 overlap between two C atoms.
- B is correct. The N atoms in **G** has the same hybridisation state as carbon in benzene (i.e. sp^2) except that in place of a C–H bond pair, a lone pair now occupies the sp^2 hybrid orbital. The unhybridised p orbital of each N atom overlap with the adjacent p orbitals of C atoms to form a delocalised ring of π electron cloud.
- C is correct. There are two sp^2 hybridised C atoms and both are bonded to sp^3 hybridised C atoms.
- D **is incorrect**. From option **C**, the two sp^2 hybridised C atoms are not bonded to any N atom. And the N atom in **H** is likely to be sp^3 hybridised.

5 $pV = nRT$

A

for the same amount of He gas, n and R remain constant throughout. He gas will diffuse to occupy both vessels J and K when the valve is opened, i.e. volume of space occupied by gas = $(v + 3v)$

$$\begin{aligned} p_i &= 1 \times 10^5 \text{ Pa} \\ T_i &= (20 + 273) \text{ K} \\ V_i &= v \end{aligned}$$

$$\begin{aligned} p_f &= ? \text{ Pa} \\ T_f &= (100 + 273) \text{ K} \\ V_f &= 4v \text{ (sum of vessels J and K)} \end{aligned}$$

$$\left(\frac{pV}{T}\right)_i = \left(\frac{pV}{T}\right)_f$$

$$\begin{aligned} p_f &= \frac{(1 \times 10^5)(v)}{(20 + 273)} \times \frac{(100 + 273)}{4v} \\ &= \underline{3.18 \times 10^5 \text{ Pa}} \end{aligned}$$

6 Property 1: melting point of chlorides

B

Chlorides of Mg form giant ionic lattice structures, have much higher melting point than simple molecular chlorides of Al, Si and P

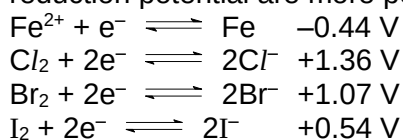
Property 2: pH of oxides

oxides	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀
Structure	giant ionic lattice		giant molecular	simple molecular
Reaction with water	weakly alkaline solution (pH $\approx$ 9)		insoluble (pH = 7)	
			acidic solution (pH = 1)	

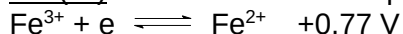
The most obvious clue to identifying the property would be the plateau from Al to Si, which is due to the insoluble oxides of the two elements.

7 On heating with Cl₂, Br₂ and I₂, the respective iron(II) halides will be formed, since their reduction potential are more positive than that of Fe²⁺/Fe.

C

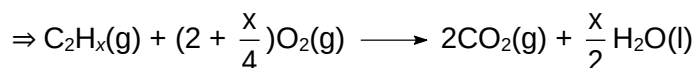


Due to the oxidising nature of the halogens, Cl₂ and Br₂ can further oxidise the iron(II) to iron(III) since their reduction potential are more positive than that of Fe³⁺/Fe²⁺.



(Note: Fe is not oxidised to Fe³⁺ in a single reaction because the reduction potential of Fe³⁺/Fe (−0.04 V) is less negative than that of Fe²⁺/Fe (−0.44 V). Fe will more likely be oxidised to Fe²⁺.)

- 8 Given that molar ratio of carbon dioxide : hydrocarbon is 2 : 1, there are two carbon atoms. $\therefore$ let formula of hydrocarbon be C_2H_x A



Since the remaining O_2 can burn up exactly 30 cm^3 of the same hydrocarbon, this means that 100 cm^3 of O_2 can burn 40 cm^3 of the hydrocarbon.

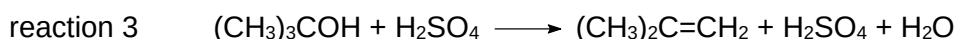
$$\Rightarrow \text{Molar ratio of hydrocarbon : } O_2 = 40 : 100 = 1 : \frac{5}{2}$$

$$\Rightarrow 2 + \frac{x}{4} = 2.5$$

$$\Rightarrow x = 2$$

Formula of hydrocarbon is C_2H_2 .

- 9 reaction 1 $Al_2O_3 + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3H_2O$ A

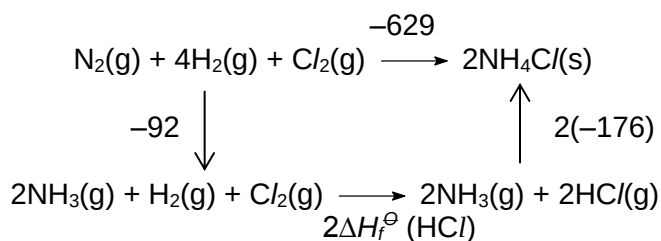
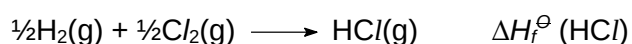


Reaction 1 is an acid–base reaction (clue: O.N. of the elements remain).

Reaction 2 is a redox reaction (most obvious clue: O.N. of Cu increases; S decreases).

Reaction 3 is an elimination reaction; H_2SO_4 is a catalyst here (clue: H_2SO_4 remains unchanged after the reaction).

- 10
$$\begin{array}{llll} N_2(g) + 3H_2(g) & \longrightarrow & 2NH_3(g) & \Delta H^\ominus = -92\text{ kJ mol}^{-1} \\ N_2(g) + 4H_2(g) + Cl_2(g) & \longrightarrow & 2NH_4Cl(s) & \Delta H^\ominus = -629\text{ kJ mol}^{-1} \\ NH_3(g) + HCl(g) & \longrightarrow & NH_4Cl(s) & \Delta H^\ominus = -176\text{ kJ mol}^{-1} \end{array}$$
 C



$$\begin{aligned} 2\Delta H_f^\ominus(HCl) &= -(-92) + (-629) - 2(-176) \\ \Delta H_f^\ominus(HCl) &= \underline{-92.5\text{ kJ mol}^{-1}} \end{aligned}$$

- 11 Info from *Data Booklet*, $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$

B

$$\begin{aligned}\text{Heat absorbed by water} &= (160 - 60) \times 4.18 \times (68 - 25) \\ &= 17974 \text{ J}\end{aligned}$$

$$\begin{aligned}\text{Heat released by ethanol} &= 1370 \times 1000 \times \frac{1.0}{2(12.0) + 6(1.0) + 16.0} \\ &= 29783 \text{ J}\end{aligned}$$

The difference in the heat transferred is due to heat loss. Hence in this case, efficiency can be calculated by taking the ratio $\frac{\text{heat absorbed}}{\text{heat released}}$.

$$\begin{aligned}\% \text{ efficiency} &= \frac{17974}{29783} \times 100\% \\ &= \underline{60\%}\end{aligned}$$

- 12 The reactions taking place in the catalytic converter include simultaneous reduction of NO_x and oxidation of CO (option C), oxidation of unburnt hydrocarbons (option D) and oxidation of CO (option B).

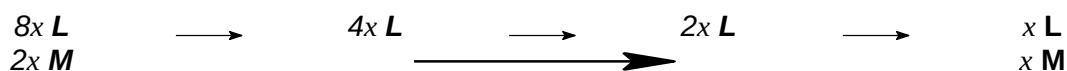
A

Hydrocarbons contain C and H atoms. Carbon is not a hydrocarbon.

- 13 The easiest approach to answering this question is to start from the *equimolar quantities of L and M* after 9 min and work backwards. Let the amount be.

C

Since L has undergone 3 half-lives, its initial concentrations will be $8L$.
Initial concentration of M will be $2M$ as it has undergone 1 half-lives.



$$\begin{aligned}\therefore \text{value of } n &= \frac{8x}{2x} \\ &= \underline{4}\end{aligned}$$



- A** is incorrect. Adding a catalyst does not affect the equilibrium yield; it only increases the rate of achieving equilibrium because the rates of both forward and backward reaction increase by the same extent.
- B** is incorrect. Adding neon at constant pressure will decrease the partial pressures of the particles present (both reactants and products)*. Since the reaction results in a decrease in no. of moles of gaseous particles, position of equilibrium (P.O.E.) shifts to the left to increase the pressure, resulting in a decrease in equilibrium yield.

$$* [p_a = \frac{n_a}{n_{\text{total}}} p_{\text{total}}, n_{\text{total}} \text{ increases while } n_a \text{ and } p_{\text{total}} \text{ remain}]$$

Addition of an inert gas at constant volume will have no impact on the P.O.E. because the partial pressure of the reactants and products are not affected.

$$[p_a = \frac{n_a}{n_{\text{total}}} p_{\text{total}}, n_{\text{total}} \text{ and } p_{\text{total}} \text{ increase to the same extent while } n_a \text{ remains}]$$

- C** is **correct**. Decreasing the temperature favours the forward exothermic reaction and P.O.E. shifts to the right (equilibrium yield increase) and the rate of production of ammonia is slower as the average kinetic energy of the particles decrease.
- D** is incorrect. Removing N_2 at constant volume ($\Rightarrow$ partial pressure of N_2 decreases) will cause the P.O.E. to shift to the left (decrease in yield) and rate of production of ammonia is slower due to a decrease in concentration of reactants.

- 15 Since $\Delta G^\ominus < 0$ B
 $\Rightarrow$ the forward reaction is spontaneous and
 $\Rightarrow$ position of equilibrium lies to the right.
 $\Rightarrow$ the amount of products at equilibrium is more than that of the reactants.
 i.e. K_p is greater than 1

Alternatively, using $\Delta G^\ominus = -RT \ln K$
 $\Rightarrow K > 1$

- 16 By inspection, ammonia is in excess. An alkaline buffer (excess unreacted NH_3 and NH_4^+) is obtained. C

$$\begin{aligned} [\text{NH}_4^+] &= n\text{HCl} \div \text{total volume} \\ &= (30 \times 0.1) \div 50 \\ &= 0.06 \text{ mol dm}^{-3} \end{aligned}$$

$$\begin{aligned} [\text{NH}_3] &= (20 \times 0.2 - 30 \times 0.1) \div 50 \\ &= 0.02 \text{ mol dm}^{-3} \end{aligned}$$

$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$$

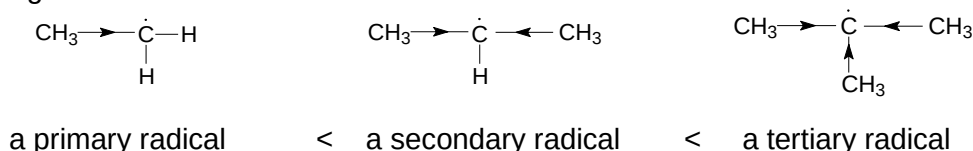
$$\begin{aligned} [\text{OH}^-] &= 6 \times 10^{-6} \\ \text{pOH} &= 5.2 \\ \text{pH} &= 8.8 \end{aligned}$$

or use

$$\text{pOH} = \text{p}K_b + \lg \frac{[\text{NH}_4^+]}{[\text{NH}_3]}$$

- 17 By inspection, the base is diprotic (2 equivalence points) and likely to be a weak one (low initial pH). And since the final pH is approximately 1, the acid added is likely a strong acid. C

- 18 Due to the inductive electron-donating effect of alkyl groups, the stability of the radicals are in increasing order of: **C**



Thus, the tertiary radical being the most stable will be formed most readily.

- 19 The following hint "... comprising of 10 linearly fused benzene ring ..." is useful in solving the question. **C**

- is correct.** Each C atom is sp² hybridised. The sp² hybridised orbitals are arranged in a trigonal planar manner about each C atom, giving a bond angle of 120°.
- is correct.** The delocalised ring of π electrons is formed by the overlapping of p orbitals on adjacent C atoms.
- is incorrect. The carbon-carbon bond energies should be intermediate between that of the bond energy of a C-C bond in alkane and the C=C bond in alkene.
[refer to the *Data Booklet* – bond energy of C=C (benzene) is 520 kJ mol⁻¹, between 350 kJ mol⁻¹ (C-C) and 610 kJ mol⁻¹ (C=C)]
- is incorrect. The valency of each C is still 4. In each C, 3 valence electrons are used in the formation of three σ bonds, and 1 electron is delocalised into the 42 π-electron system.

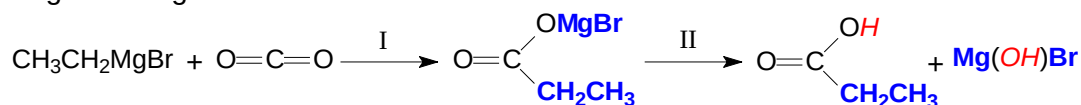
- 20 Chlorofluoroalkanes generate Cl• radicals which destroy stratospheric ozone. The more Cl atoms that is present in the molecule, the more Cl• radicals can be generated, and the higher the ODP. **A**

- 21
$$\text{C}_2\text{H}_5\text{MgBr} + >\text{C}=\text{O} \xrightarrow{\text{I}} \begin{array}{c} \text{O} \text{MgBr} \\ | \\ >\text{C} \\ | \\ \text{C}_2\text{H}_5 \end{array} \xrightarrow{\text{II}} \begin{array}{c} \text{OH} \\ | \\ >\text{C} \\ | \\ \text{C}_2\text{H}_5 \end{array} + \text{Mg}(\text{OH})\text{Br}$$
 D

Step I involves the addition of **-C₂H₅** and **-MgBr** across the carbonyl group (in bold). As the reactive carbon is electron deficient, it is susceptible to nucleophilic attack.

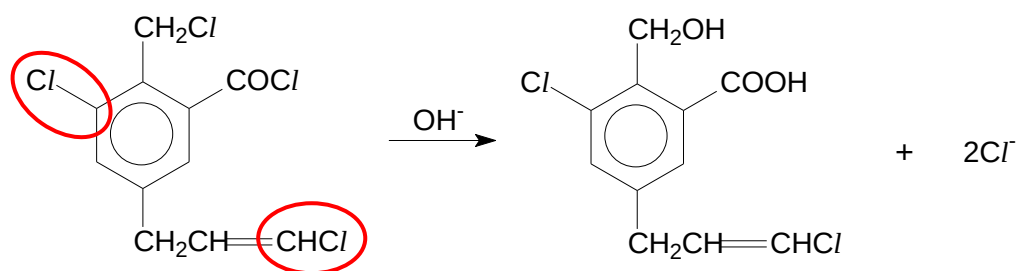
Step II involves the hydrolysis of O-Mg bond (observe the **H** and **OH** added – in italics). This reaction is dependent on the polarity across the O(δ-) – Mg(δ+) bond.

- 22 **1 is correct.** Similar to the above steps, **-C₆H₅** and **-MgBr** are added to the carbonyl group of CH₃CHO, followed by hydrolysis of the O-Mg bond to form **C₆H₅CH(OH)CH₃** and **Mg(OH)Br**. **D**
- 2 is incorrect. The reactant has a longer carbon chain (additional -CH₂-).
- 3 **is correct.** By observation, a carboxylic acid will be formed when CO₂ reacts with a Grignard reagent.



(To visualize, replace the 2 R group substituents on the reactive carbon (>C) with an oxygen doubly bonded (=O) to the carbon in CO₂.)

23



B

Nucleophilic substitution does not occur due to the delocalisation of the lone pair of electrons on the Cl atom into the benzene ring (or adjacent C=C double bond), creating a partial double bond character in the C–Cl bond (circled). The polarity of the C–Cl bond is also decreased due to the delocalisation and hence the C atom of the C–Cl bond is less electron deficient and less susceptible to attack by the OH[−] nucleophile.

Steric hindrance due to repulsion of the nucleophile by the π electron cloud could also be used as a reason.

24

- 1 is incorrect. Absence of CH₃COR or CH₃CH(OH)R. Hence, it does not react with I₂/OH[−](aq).
- 2 is incorrect. All three 2° alcohol present in *Indian Yellow* will be oxidised by Cr₂O₇^{2−}/H⁺ to give a compound with four carbonyl groups.
- 3 is incorrect. Only 2 atoms of Br will be substituted on the benzene ring at the 2nd and 4th position with respect to phenol group. Even if you had use the concept that ether (R–O–R group) has similar reactivity as phenol, still the number of Br atom incorporated will be less than 6 Br atoms. However, do note that reactivity of ether is not in the syllabus. Hence, you have to follow the clue given in the question to decide when it is reactive and when it is assumed to be inert. (refer to N2008/III/5d, N2013/II/3aiv, N2009/II/24 etc)
- 4 **is correct**. Each mole of –OH group reacts with Na metal to form $\frac{1}{2}$ mol of H₂(g). Since there are 5 mol of –OH group per mol of *Indian Yellow*, 2.5 mol of H₂(g) is formed.

D

25

decreasing pK_a ⇒ increasing K_a and increasing strength of acid

Observe that all four compounds are carboxylic acids, hence the order of strength is dependent on the substituent present and its position relative to –COOH group.

A

Recall: Strength of an acid depends on the stability of its conjugate base.

(Why? pK_a + pK_b = 14 at 298 K, so the bigger the pK_b is, the smaller the pK_a will be. A stable base will mean it is unreactive/weak base ⇒ greater K_a for its conjugate acid.)

Compare carboxylate ions: **H**–C₆H₄COO[−] and **CH₃**–C₆H₄COO[−]

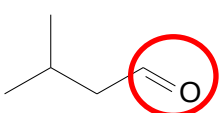
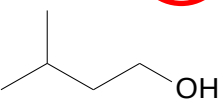
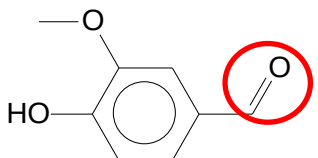
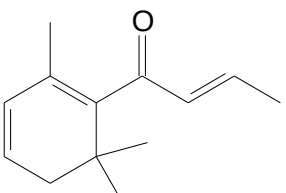
- Presence of electron donating alkyl group in CH₃C₆H₄CO₂H reduces the dispersal of the –ve charge to a smaller extent and thus stabilises the conjugate base to a smaller extent.
- Hence CH₃C₆H₄CO₂H has a larger pK_a than C₆H₅CO₂H. (narrow down to options **A** and **D**)

Compare: 4–hydroxybenzoic acid and 2–hydroxybenzoic acid.

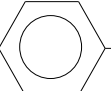
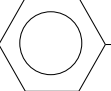
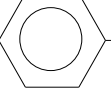
- Close proximity of the phenol group to carboxylic acid group in 2–hydroxybenzoic acid enables it to form intramolecular hydrogen bonding which helps to stabilise the conjugate base to a greater extent.
- Hence 2–hydroxybenzoic acid has a lower pK_a than 4–hydroxybenzoic acid. (option **A**)

- 26 $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$ is able to oxidise 1°, 2° alcohols and aldehyde.
 NaBH_4 is able to reduce aldehyde and ketones.

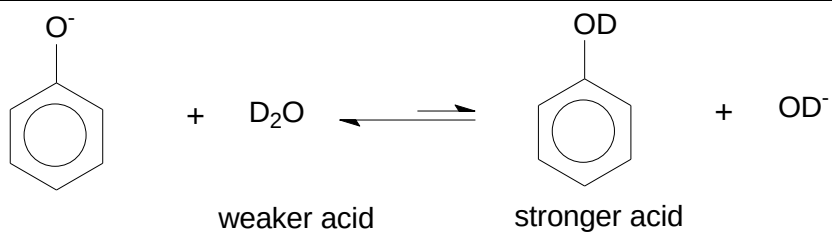
B

		$\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$	NaBH_4
1		✓	✓
2		✓	X
3		✓	✓
4		✓	X

27

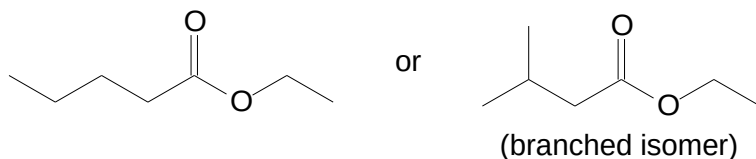
	Type of reaction	Products
A	base-catalysed hydrolysis	 $\text{CO}_2^- \text{Na}^+ + \text{CH}_3\text{OD}$
B	base-catalysed hydrolysis	 $\text{O}^- \text{Na}^+ + \text{CH}_3\text{CO}_2^- \text{Na}^+$
C	acid-catalysed hydrolysis	 $\text{OD} + \text{CH}_3\text{CO}_2\text{D}$
D	no reaction	 $\text{O}^- \text{Na}^+$

C

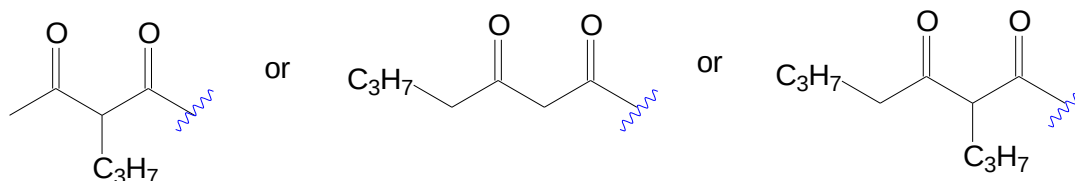


Since $\text{C}_6\text{H}_5\text{OD}$ is a stronger acid than D_2O , the position of equilibrium lies to the left as $\text{C}_6\text{H}_5\text{OD}$ is a better proton donor. $\text{C}_6\text{H}_5\text{O}^-$ will not react with D_2O to give a good yield of $\text{C}_6\text{H}_5\text{OD}$.

- 28 Given that the first member of the series is ethyl ethanoate
 $\Rightarrow n = 0$
 $\therefore$ the fourth member of the series ($n = 3$) will be either



By observation, a possible product must have one of the following structures.

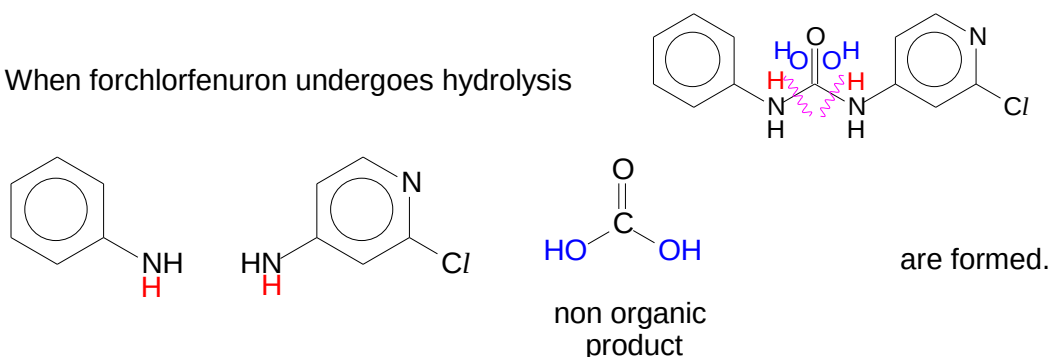


Hence,

- A** is incorrect. One of its reactants is the fifth member of the series (i.e. $n = 4$).
B **is correct.** It is formed between two fourth members of the series.
C is incorrect. It is formed between a second and a third member of the series.
D is incorrect. It is not part of the ethyl ester series. It is a butyl ester.

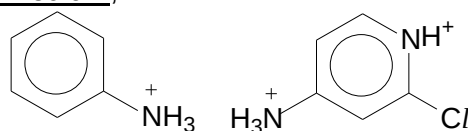
29

When forchlorfenuron undergoes hydrolysis



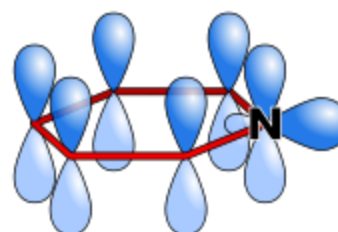
However as the reaction was carried out in acidic medium,

the basic groups will be protonated (e.g. $-\text{NH}_2$)



FYI

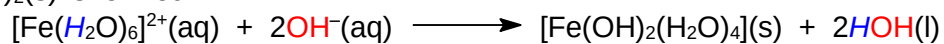
The N atom in the pyridine is basic as its lone pair is not delocalised in the ring but occupies the sp^2 hybrid orbital. Instead, it is the unhybridised p orbital (with an electron) overlap with the p orbitals of the adjacent C atoms to form an aromatic ring as illustrated in the diagram on the right.



B

D

- 30** When excess of NaOH (aq) are added to Fe(NO₃)₂(aq), a dirty green precipitate of Fe(OH)₂(s) is formed. **B**



OH⁻ accepts a proton from the hydrated iron(II) complex (acid–base reaction), resulting in the precipitation of Fe(OH)₂(s).

On standing in air, Fe(OH)₂(s) is oxidised to Fe(OH)₃(s), reddish brown precipitate.